Surgical Endoscopy (2025) 39:5879–5888
https://doi.org/10.1007/s00464-025-12018-3

External validation of a motion capture-based surgical skill assessment system in laparoscopic simulation training environments

Koki Ebina[2] · Takashige Abe[1] · Kiyohiko Hotta[1] · Yan Lingbo[2] · Chihiro Kamijo[1] · Madoka Higuchi[1] · Masafumi Kon[1] · Hiroshi Kikuchi[1] · Haruka Miyata[1] · Ryuji Matsumoto[1] · Takahiro Osawa[1] · Sachiyo Murai[1] · Taku Senoo[2] · Shunsuke Komizunai[3] · Teppei Tsujita[4] · Kazuya Sase[5] · Xiaoshuai Chen[6] · Atsushi Konno[2] · Nobuo Shinohara[1]

Received: 1 May 2025 / Accepted: 13 July 2025 / Published online: 26 July 2025
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Abstract

Purpose To externally validate our surgical skill assessment system, which provides comprehensive real-time feedback based on motion capture (Mocap) metrics of laparoscopic instruments in simulation training environments.

Methods Participants conducted periaortic tissue dissection (Dissection Task) and parenchymal closure of the kidney (Suturing Task) using porcine cadaveric organs. Surgical instrument movements were measured by the MoCap system, and all training sessions were video-recorded. Utilizing our previously developed machine-learning (ML) algorithm, participants received quantitative feedback based on their motion features immediately after the training, including three-tier classification results, which meant their skill levels being consistent with those with $0 \leq$ to ≤ 9 surgical caseloads of laparoscopy (novice), $10 \leq$ to ≤ 49 (intermediate), and ≥ 50 (experienced), and predicted Global Operative Assessment of Laparoscopic Skills (GOALS) scores. Automatically generated feedback comments were also provided through explainable artificial intelligence technology, and participants' satisfaction with this skill assessment system was also recorded. For the three-tier classifications, accuracy was calculated by comparison between estimated and actual surgical experience levels. The correlation and mean absolute error (MAE) were calculated between the algorithm-predicted and expert-evaluated mean GOALS scores.

Results A total of 38 urologists, 4 junior residents, and 10 medical students voluntarily participated in the current training. Several participants repeatedly performed the training, and so 107 datasets were included in the present analysis. Validation analyses demonstrated overall classification accuracies of 67.3 and 56.9% for Dissection Task and Suturing Task, respectively, when excluding the subsequent data of repeaters. In terms of GOALS score prediction, the correlation coefficient and MAE were 0.86 and 2.51 ± 1.82 for Dissection Task, and 0.91 and 1.56 ± 1.20 for Suturing Task, respectively, when compared with experts' evaluations. Overall, 88% of the participants found the feedback easy to understand, and 75% of participants were satisfied with the evaluation.

Conclusions Our real-time feedback system was viewed favorably by the participants, and the ML algorithm was a reliable method, especially the predicted GOALS score. This system could facilitate objective performance assessment, enhance simulation-based training efficacy, and potentially accelerate surgical skill acquisition.

Keywords Laparoscopic surgery · Simulation training · Motion capture · Surgical education · Skill evaluation

✉ Takashige Abe
takataka@rf6.so-net.ne.jp

[1] Department of Renal and Genitourinary Surgery, Faculty of Medicine, Hokkaido University, North-15, West-7, North Ward, Sapporo 060-8638, Japan

[2] Graduate School of Information Science and Technology, Hokkaido University, Sapporo, Japan

[3] Faculty of Engineering and Design, Kagawa University, Takamatsu, Japan

[4] Department of Mechanical Engineering, National Defense Academy of Japan, Yokosuka, Japan

[5] Department of Mechanical Engineering and Intelligent Systems, Tohoku Gakuin University, Sendai, Japan

[6] Graduate School of Science and Technology, Hirosaki University, Hirosaki, Japan

Laparoscopic surgery has been widespread due to its distinctive advantages: magnified visualization of anatomical structures, minimal invasiveness, and rapid postoperative recovery. However, surgeons need to acquire proficiency with specific surgical instruments, ambidextrous coordination of the hands, and depth perception on viewing a surgical monitor. Therefore, several studies have reported objective skill analysis of laparoscopic surgery by measuring surgical instrument movements using motion capture and other motion tracking systems [1–3]. In addition, there are several studies that have developed evaluation models that classify based on the number of surgical procedures performed using machine-learning models for the purpose of providing feedback to trainers, as well as regression models that predict scores for skill assessment indicators such as OSATS [4–6]. However, most of them were related to simple tasks such as forceps handling in dry labs, and did not take into account more complex tasks in wet labs. Furthermore, the model accuracy of the evaluation has not been externally verified, and the questionnaire survey on the acceptability of the evaluation is insufficient. Since 2017, in order to improve trainees' fundamental laparoscopic skills, we have regularly held wet-lab simulation training for laparoscopic surgeries, using porcine cadaveric organs, which consisted of periaortic tissue dissection (Dissection Task) and renal parenchymal suturing (Suturing Task) [7]. Subsequently, in order to clarify experts' surgical techniques, we developed a motion capture (MoCap)-based system to analyze their movements. Our observations revealed that experienced surgeons (≥ 50 laparoscopic cases) had shorter efficiency-related parameters such as path length and angular length, faster speed-related parameters such as average velocity/acceleration/jerk in both tasks, and improved depth perception and bimanual dexterity [8]. In addition, in order to enrich feedback for trainees, we developed two skill assessment methods using motion metrics and a machine-learning (ML) algorithm: (1) a three-tier classification model, which meant their skill levels being consistent with those with $0 \leq$ to ≤ 9 surgical caseloads of laparoscopy (novice), $10 \leq$ to < 49 (intermediate), and ≥ 50 (experienced) [9], and (2) a regression model predicting the Global Operative Assessment of Laparoscopic Skills (GOALS) score [10]. We previously reported that the median accuracy of the three-tier classification model was 68.6 and 58.4% for Dissection Task and Suturing Task [9], respectively, while the regression model demonstrated median of mean absolute error (MAE) values of 2.23 and 1.27, respectively [10]. Using this ML models, we created a real-time skill assessment system, in which participants received automated skill evaluation immediately after the training [11]. Utilizing SHAP, an explainable AI methodology [12], participants received comprehensive feedback regarding their estimated GOALS scores, delineating the positive and negative contributions of specific motion metrics to their performance evaluation. However, the accuracy of this ML algorithm has only been verified by cross validation with a limited sample size, and the accuracy of our real-time skill assessment system constructed using all data as the training set has not been verified. Additionally, there was no user-based evaluation of the comprehensive skill feedback provided by the system, and users' impressions of the feedback were unclear.

The present study had dual objectives: primarily, to perform external validation of our ML algorithm through additional data collection to verify the accuracy of our skill assessment system; secondarily, to elucidate participants' perspectives regarding our novel skill assessment system.

Material and methods

The Ethical Review Board for Life Science and Medical Research of Hokkaido University Hospital approved this study (No. 018-0257). Participants provided written informed consent for the research.

Training task and measurement system

Figure 1A shows an overview of the experimental environment. We previously reported our wet-lab training for laparoscopic surgery [8]. Briefly, cadaveric porcine organs were placed in a box trainer (Endowork ProII®, Kyoto Kagaku, Japan) and one of the authors (TA or CK) served as a scopist using the endoscopic video system (VISERA Pro Video System Center OTV-S7Pro, Olympus, Japan). In Dissection Task, participants performed periaortic tissue dissection, dividing the encountered the mesenteric vessels after applying Hem-o-lock (Fig. 1B). In Suturing Task, using two 15-cm 2–0 CT-1 VICRYL® threads, participants tied three square single knots at two sites on a kidney (Fig. 1C). All training sessions were digitally video-recorded for subsequent analysis. The construct validity of the training has already been reported [7]. The participants of the training sessions were recruited using the snowball sampling method.

Instrumental motion metrics were measured utilizing the optical Mocap system (OptiTrack V120: Trio, NaturalPoint Inc., USA). Mocap is a system that uses multiple infrared cameras to measure the three-dimensional position of infrared-reflective markers based on the principle of triangulation. By using multiple markers, it is possible to measure the position and orientation of rigid bodies in real time. In this study, four infrared-reflective markers affixed to the surgical instruments, enabling precise measurement of forceps tip positioning (X, Y, and Z coordinates) and spatial orientation (Roll, Pitch, and Yaw). The system had a simple configuration with no need for dynamic calibration, making it easy to introduce into an experimental environment. Each marker

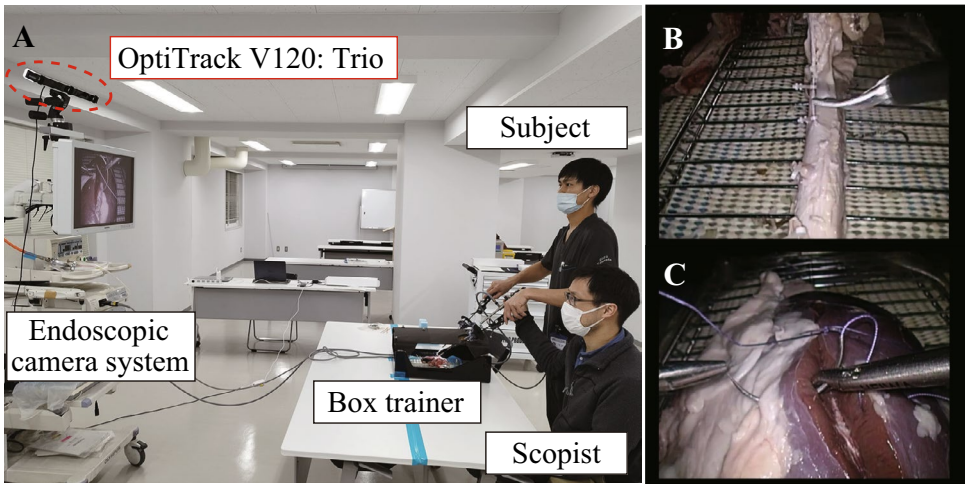


Fig. 1 An overview of the experimental environment. **A** An overview of the experimental environment. The movements of surgical instruments were measured by the MoCap system. **B** Appearance of the porcine aorta used in the training. **C** Appearance of the porcine kidney parenchyma used in the training

set had a unique placement pattern, and so the system could measure multiple surgical instruments simultaneously. To reduce the effect of noise, the Savitzky–Golay filter [13], a data-smoothing method, was applied to the forceps tip position and orientation data, and derivatives of the tip position (velocity, acceleration, and jerk) were also calculated by the filter. The measurement frequency was set to 120 Hz.

Skill assessment system

Our prior publication described the contents of our skill assessment system [11]. The system quantifies instrumental kinematic parameters, including mean velocity, acceleration, jerk, path length, and bimanual dexterity, and two skill assessment models using machine-learning (ML) algorithms are built-in. The primary algorithm classifies surgical competency into three hierarchical tiers corresponding to procedural laparoscopic experience: novice (0–9 laparoscopic procedures), intermediate (10–49 procedures), and experienced (≥ 50 procedures). The secondary algorithm generates predictive GOALS scores derived from motion analysis. The classification model utilizes Principal Component Analysis-based Support Vector Machine (PCASVM), while the GOALS prediction model employs Principal Component Analysis-based Support Vector Regression (PCASVR). These two models were selected because our previous study confirmed that they have the highest accuracies among several machine-learning algorithms through nested and repeated tenfold cross validation [11]. In this study, we constructed ML models using the above mentioned ML algorithms, and all data utilized for cross validation in our previous studies were used as training data ($n = 104$). Therefore, the data collected in this study were only used as a test set for accuracy verification, not for model construction. In addition, the model was constructed for each of the Dissection and Suturing Tasks and was evaluated using the same tasks in the verification.

To present the background basis for evaluation, SHAP, one of the explainable AI methods [12], was also installed. SHAP elucidates how the features contribute to a prediction result by approximating the Shapley value in cooperative game theory. By calculating the contribution of each feature to the individual prediction result, it can provide the trainee with positive and negative points during the training session. Figure 2 shows an example screenshot of the evaluation results in Dissection Task. The evaluation items are as follows:

(a) Radar chart of GOALS scores
(b) Result of three-tier classification
(c) SHAP-derived feedback comments
(d) Button to check the waterfall plot of SHAP
(e) Principal component scores
(f) Raw kinematic parameters

Component (a) is a radar chart of the GOALS score estimated from the regression model (blue area). The chart also displays the median GOALS scores for subjects with the same experience level (orange area). Component (b) is the estimated skill level based on a three-tier classification model (novice, intermediate, or experienced). Component (c) is the feedback comment based on the calculated SHAP values, with (d) offering detailed contribution visualization. Component (e) presents principal component scores achieving a minimum of 75% cumulative variance explanation, and the scores of subjects with the corresponding category in the original data are also displayed. Component (f) shows the raw values of kinematic metrics for each instrument. Figure 3A shows an example of the water fall plot in the tissue-handling domain of GOALS. The metrics were placed in descending order of contribution, with positive and negative influences denoted in red and blue, respectively. Figure 3B shows an example of the feedback comments regarding tissue handling, which consists of the three elements regarding

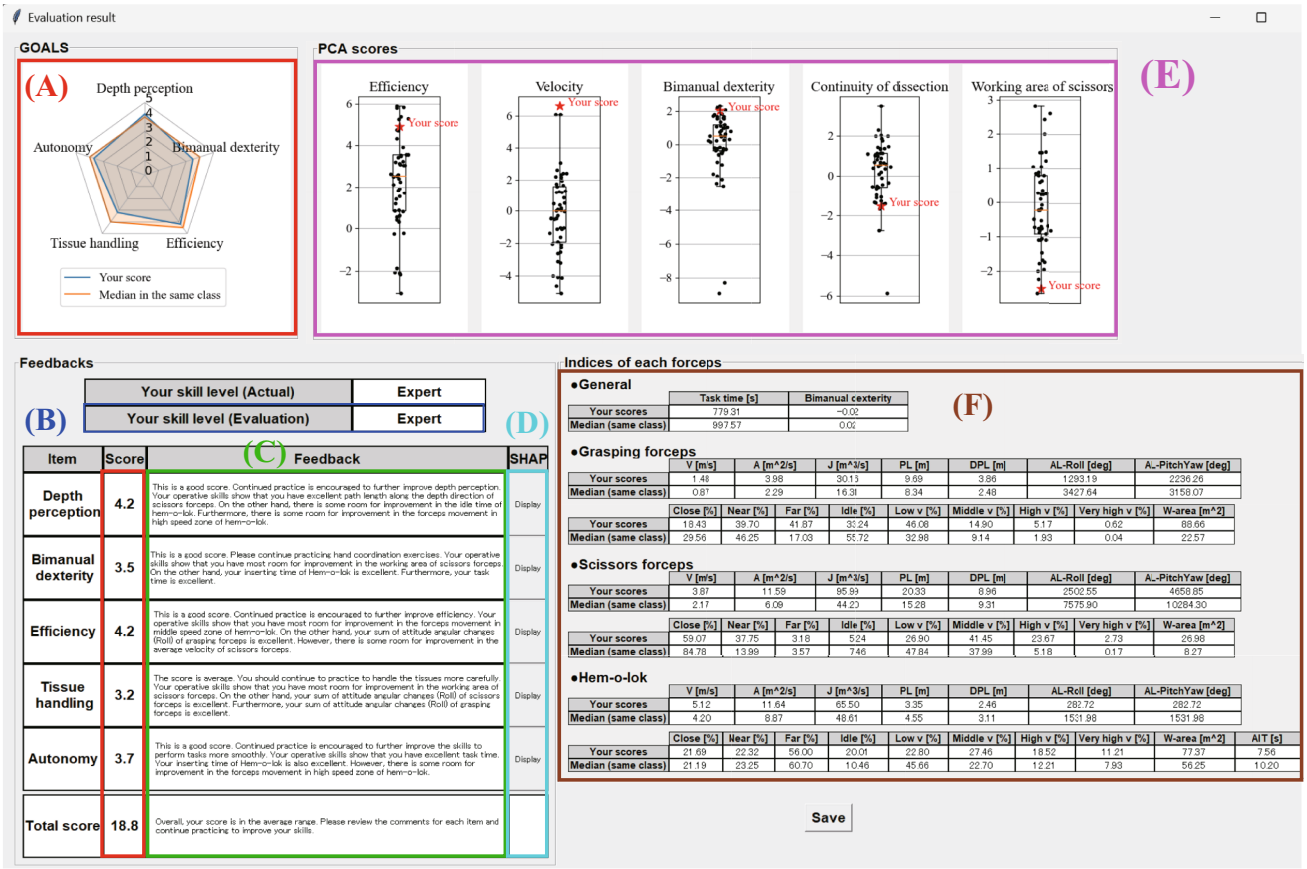


Fig. 2 Example screenshot of the evaluation results in Dissection Task. Component **A** is a radar chart of the GOALS score, **B** is the estimated skill level based on a three-class classification model, **C** is the feedback comment based on the calculated SHAP values, with **D** offering detailed contribution visualization. Component **E** presents principal component scores, and **F** shows the raw values of kinematic metrics for each instrument

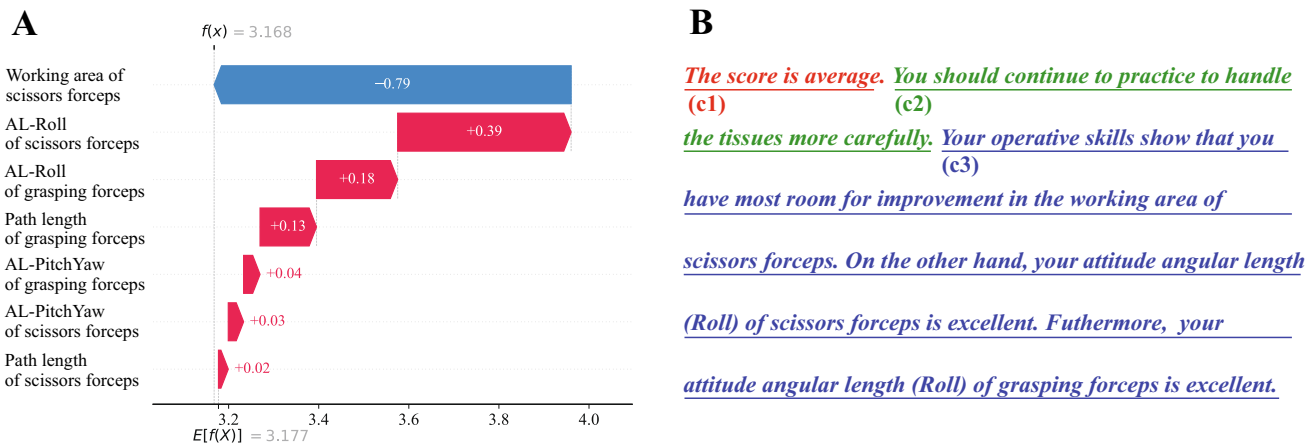


Fig. 3 Feedback based on SHAP. **A**: An example water fall plot in the tissue-handling domain of GOALS. The metrics were placed in descending order of contribution, with positive and negative influences denoted in red and blue respectively. **B**: An example of the feedback comments regarding tissue handling. The comment consists of three elements: (c1) the estimated score, (c2) the overall evaluation, and (c3) the top three indices contributing to SHAP

(c1) the estimated score, (c2) the overall evaluation, and (c3) the top three contributing indices in SHAP. All feedback comments were generated automatically by combining pre-registered comment sets prepared by discussion between two

authors (KE and TA) based on estimated GOALS scores and contributions of kinematic indices obtained from SHAP analysis.

Post-training questionnaire survey

After the training, we collected demographic data from the participants, and a questionnaire survey was also conducted to verify assessment validity. The questions were as follows:

Question 1 (Q1): Do you think the automated feedback results were easy to understand?

Question 2 (Q2): Is your predicted skill level acceptable?

Question 3 (Q3): In the current skill assessment system, which items do you consider to be the most helpful?

Question 4 (Q4): Vice versa, which item do you consider to be the least helpful?

Responses to Q1 and Q2 utilized a 5-point Likert scale (1: strongly disagree; 2: disagree; 3: neutral; 4: agree; 5: strongly agree), and those to Q3 and Q4 employed multiple choice. For subjects participating in multiple sessions, only initial response data were incorporated into the analyses. For Q3 and Q4, the responses were quantified as + 1 (most helpful), − 1 (least helpful), or 0 (neutral) if neither was selected, and the mean response value was then calculated.

Movie assessments

Two experts (TA and KH) independently evaluated the anonymized movies according to the GOALS formula [14]. The mean scores were used for subsequent analyses. In Suturing Task, "Tissue Handling" score was not evaluated, because needle and thread manipulation constituted the main component of this task. Therefore, Dissection Task was scored on a 5 to 25-point scale, while Suturing Task was scored on a 4- to 20-point scale.

External validation protocol

As mentioned earlier, the purpose of this study was to verify the accuracy of our real-time skill assessment system through external validation. Three-tier classification accuracy was determined through comparative analysis of predicted versus actual procedural laparoscopic experience categories. The accuracy of the evaluation was calculated by excluding the subsequent data of repeaters, because it becomes inaccurate to evaluate the accuracy of the evaluation if subjects, especially novices, could improve their skills after multiple training sessions. For the GOALS regression model, the accuracy was assessed using MAE between the predicted and mean GOALS scores evaluated by the two experts. In GOALS-based evaluation, unlike the three-tier evaluation based on the number of surgical procedures, data from duplicate participants are anonymized and evaluated independently for each session, so there is no impact on the verification results of the accuracy of the evaluation model. Therefore, we verified the accuracy of the GOALS-based evaluation based on all data.

Statistics

Statistical analysis was performed using JMP Pro 17 (SAS, Japan), in order to calculate the intra-class correlation coefficient for GOALS scores obtained by human raters, and assess questionnaire results by calculating 95% confidence intervals and basic statistics. Regarding the class classification accuracy and MAE, Excel® was utilized.

Results

Table 1 summarizes the participants' backgrounds. Thirty-eight urologists, 4 junior residents, and 10 medical students participated in the training. Twenty-one subjects participated in multiple training sessions, yielding 107 Dissection Task and 109 Suturing Task sessions for external validation study. Excluding two datasets due to recording failure in Suturing Task, 107 novel datasets were incorporated into the validation protocol. Supplementary Table 1 shows the distribution of task sessions among the 52 participants.

Three-class classification performance analysis

Figure 4 illustrates the classification confusion matrix, and Table 2 shows the performance metrics of the classification model. The overall classification accuracy (micro average of accuracy) after excluding the subsequent data of repeaters was 67.3% (95% CI: 61.3–73.3%) for Dissection Task and 56.9% (95% CI: 50.1–63.7%) for Suturing Task.

Table 1 Participants' backgrounds

	$n = 52$
Background	Urologic surgeon, $n = 38$
	Junior resident, $n = 4$
	Medical student, $n = 10$
Age, years	Median 33 (range, 22–57)
Sex	Male/Female = 42/10
Experience of laparoscopic surgery	0–9, $n = 22$
	10–49, $n = 13$
	50–99, $n = 5$
	100–499, $n = 10$
	≥ 500, $n = 2$
Dominant hand	Right/left = 50/2

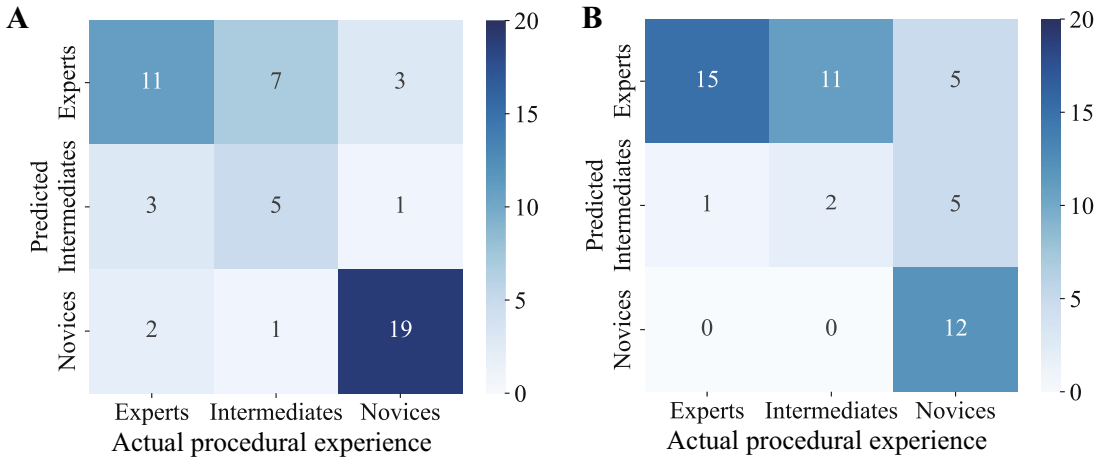


Fig. 4 The classification confusion matrix (**A**: Dissection Task, **B**: Suturing Task). The X-axis shows the actual procedural experience, Y-axis shows the predicted values, and numbers on the matrix indicate the numbers of data

Table 2 Performance metrics of classification model

	Experienced	Intermediate	Novice	Macro average	Micro average
(a) Dissection Task					
Accuracy	0.712	0.769	0.865	0.782	0.673
Precision	0.524	0.556	0.864	0.648	
Recall	0.688	0.385	0.826	0.633	
F1 score	0.595	0.455	0.844	0.631	
(b) Suturing Task					
Accuracy	0.667	0.667	0.804	0.712	0.569
Precision	0.484	0.250	1.000	0.578	
Recall	0.938	0.154	0.545	0.546	
F1 score	0.638	0.190	0.706	0.512	

GOALS score prediction model performance

Figures 5A and 5B show a scatter plot of GOALS scores rated by the two experts, and Figs. 5C and 5D show a Bland–Altman plot. The intra-class correlation coefficient was 0.83 for Dissection Task, and 0.81 for Suturing Task. Figure 6 presents scatter plots between the Mocap-derived GOALS score predictions (Y-axis) and mean score of two experts (X-axis) for both tasks (6A: Dissection, 6B: Suturing). The correlation coefficient and MAE were 0.86 and 2.51 ± 1.82 (95% CI: 2.16–2.86) for Dissection Task and 0.91 and 1.56 ± 1.20 (95% CI: 1.40–1.92) for Suturing Task.

Validation of platform usability

Figure 7A presents questionnaire responses regarding feedback comprehensibility (Q1) and skill classification acceptability (Q2). Mean response values were 4.08 (95% CI: 3.90–4.25) for Q1 and 3.88 (95% CI: 3.65–4.12) for Q2. Figure 7B illustrates comparative utility assessment of system components (Q3: most helpful, Q4: least helpful). GOALS score feedback demonstrated superior utility, while three-tier classification and raw metric values demonstrated reduced informative value, with 95% confidence intervals extending below zero.

Discussion

Aiming to integrate a haptic sense of authentic tissue manipulation in wet-lab training with objective metrics provided by a VR simulator, and shorten the learning curve, we have advanced the Mocap-based surgical skill assessment system in laparoscopic wet-lab training, using cadaveric porcine organs, infrared cameras, and ML algorithms [8–10]. Several researchers have also documented analogous investigations in laparoscopic simulation training. For example, Oquendo

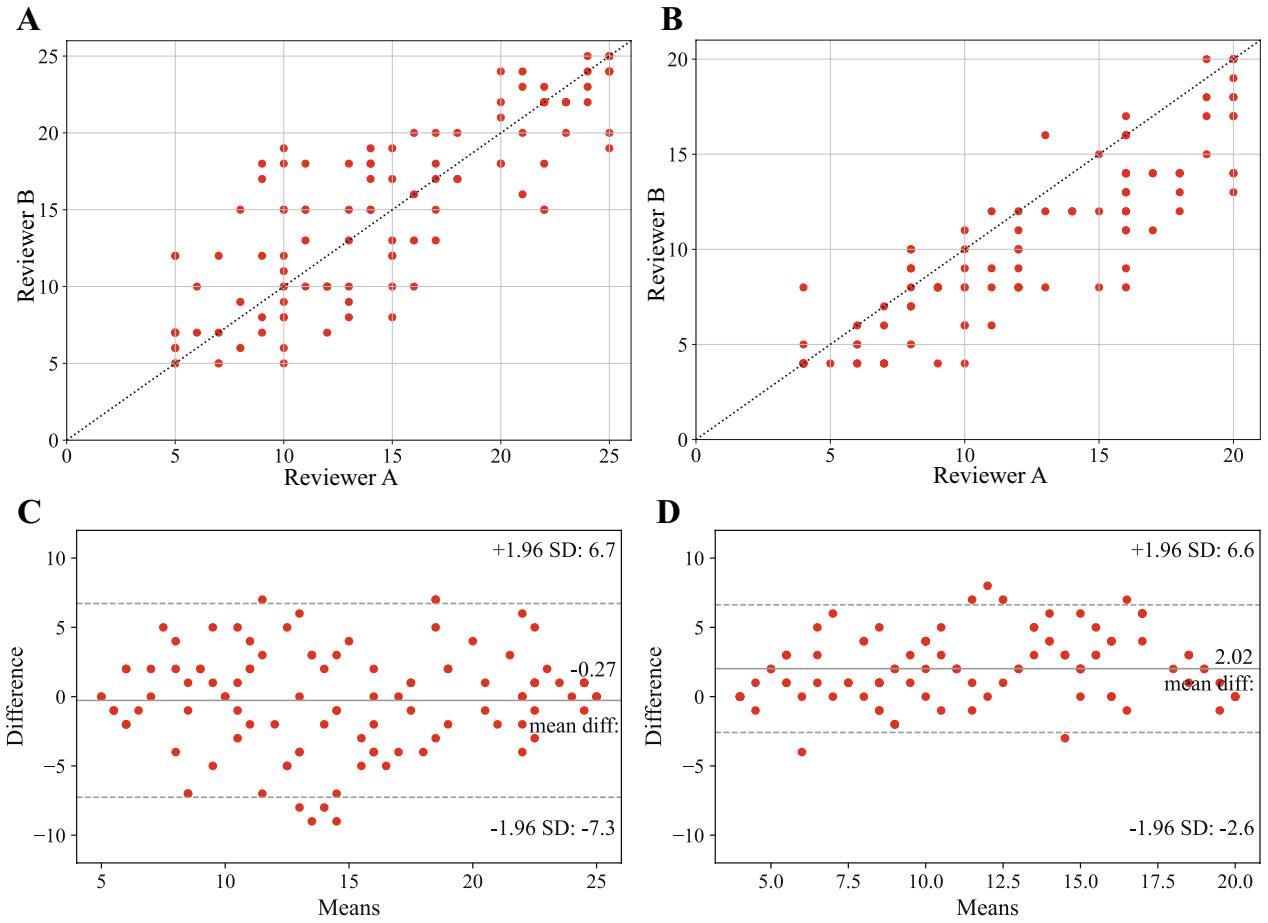


Fig. 5 Scatter plot and Bland–Altman plot of GOALS scores rated by the two experts. Scatter plot shows GOALS scores (**A:** Dissection Task, **B**: Suturing Task). The x-axis shows the scoring results of Reviewer A and the y-axis shows those of Reviewer B. Bland–Altman plot shows the difference of GOALS scores (**C**: Dissection Task, **D**: Suturing Task). The x-axis shows the mean scores and the y-axis shows the score difference

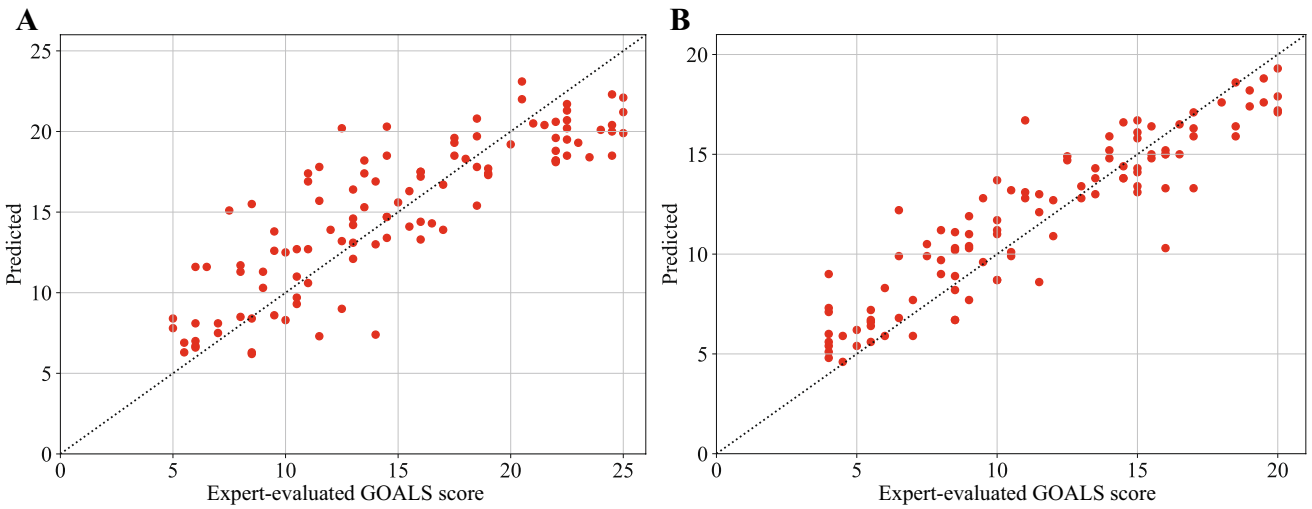


Fig. 6 The scatter plots between the Mocap-derived GOALS score predictions and the mean score of two experts (**A**: Dissection Task, **B**: Suturing Task). The X-axis shows the expert-evaluated GOALS score, and Y-axis shows the predicted values

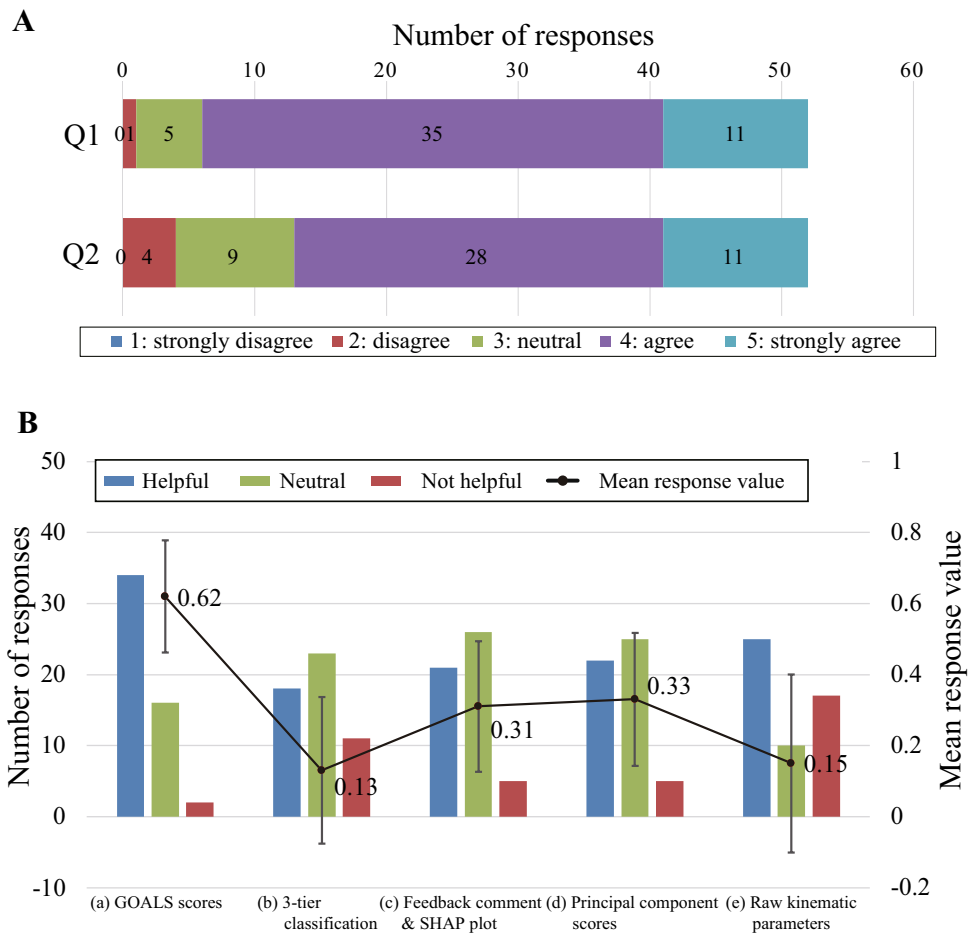


Fig. 7 Results of the questionnaire survey. **A**: Questionnaire responses regarding feedback comprehensibility (Q1) and skill classification acceptability (Q2). **B**: Questionnaire responses regarding comparative utility assessment of system components (Q3: most helpful, Q4: least helpful). The bars indicate the number of responses, line indicates the mean response value, and error bars indicate 95% confidence intervals

YA et al. demonstrated that through the implementation of a magnetic sensing system, motion data predicted Objective Structured Assessment of Technical Skill (OSATS) scores within a two-point margin, achieving 71% concordance with experts' evaluations of intracorporeal suturing training using a box trainer [4]. Additionally, Buescher JF et al. conducted a prospective randomized trial examining the education benefit of a novel box trainer system that quantified motion parameters, including path length, number of movements, velocity, and latency periods. Their findings revealed that iterative motion feedback after simulation training (bead transfer, pattern cutting, or intracorporeal knot in a novel box trainer) significantly improved several skill aspects (operative time, path length, and number of movements of left and right instruments) on conducting a laparoscopic cholecystectomy procedure using a LAP Mentor II virtual reality simulator, compared with conventional box training [15]. Compared with these previous studies, our novel training system offers distinct advantages, incorporating authentic animal tissues (porcine organs), comprehensive kinematic feedback derived from all the surgical instruments utilized, and performance assessment (a three-tier classification and estimated GOALS scoring) immediately after the training, underpinned by explainable AI methodology (SHAP). The present investigation encompassed new data acquisition and external validation of our ML algorithms. To our knowledge, our skill assessment system represents the most rigorously validated skill assessment platform for laparoscopic wet-lab simulation training.

The 3-tier classification model (stratifying participants into novice, intermediate, and experienced surgeon categories based on procedural laparoscopic experience) demonstrated accuracy rates of 67.3% for Dissection Task and 56.9% for Suturing Task being similar to our previous performance outcomes (Task 1: 68.6%, Task 3: 58.4%) [9]. As shown in Fig. 4B, for example, there were marked misclassifications between the intermediates and experienced surgeons in Suturing Task. The recall of the intermediates in this task was 0.154, as shown in Table 2. This indicates that most intermediates were not correctly identified. A similar trend was observed in Dissection task, suggesting that distinguishing intermediates is difficult. These findings underscore the limitation of utilizing the previous procedural volume as a surrogate indicator for skill proficiency, particularly within simulation training environments. The relative simplicity of simulation training tasks compared with

clinical surgical practice may obscure the differentiation between intermediate and experienced surgeon performance outcomes. Chmarra MK et al. also previously reported the limited accuracy of 74% of the three-tier classification based on the surgeon's surgical experience in dry-lab simple training tasks using a box trainee with a built-in motion tracking system [5], although their outcome was better than ours. This suggests that there are limitations to evaluating technical proficiency based solely on the number of surgical procedures performed, and that it may be more appropriate to use only the GOALS described below for evaluating skills in future training.

Regarding our GOALS regression model validation, we observed MAEs of 2.51 and 1.56 between predicted and expert-evaluated mean GOALS scores for Dissection Task and Suturing Task, respectively, closely aligning with our initial findings (median MAE: 2.24 for Dissection Task, and 1.27 for Suturing Task) [10]. Employing Myo armbands (Thalmic Labs™, Ontario, Canada) to capture acceleration, angular velocity, orientation, and Euler parameters, and ML algorithms, Kowalewski KF et al. also reported promising outcomes in automated skill assessment, with estimated OSTAS scores yielding MAE of 3.7 in the dry-lab knot-tying task [6]. These observations suggest that the integration of expert video analysis with motion-based metrics and ML algorithms constitutes an optimal approach to automated surgical skill assessment.

By leveraging SHAP (a type of explainable AI method), our novel skill assessment system provided comprehensive justification for estimated GOALS scores. As shown in Fig. 7, 88% (46/52) of the participants found the feedback easy to understand, and 75% (39/52) satisfied with the assessment outcomes. Among feedback comments, estimated GOALS scores emerged as the most informative feedback, followed by principal component scores and SHAP-derived commentary. Conversely, three-tier classification and raw metrics values were deemed less instructive by participants. We hypothesize that the quantitativeness of estimated GOALS scores and principal component scores, and the elucidatory nature of SHAP-based feedback comments contributed to participants' favorable views, whereas the relative opacity of three-tier classification and raw metrics values without any explanation reduced their perceived utility.

Limitations of this study include its small sample size, and inherent variability in porcine organs, such as the number of mesenteric vessels. As shown in Materials and Methods, we collected 107 datasets from 52 surgeons and medical students. This sample size may be too small to adequately verify the accuracy of ML models. This was due to limited resources, as the GOALS score was manually scored by expert urologists. While experts' evaluation of GOALS scores via anonymized recordings attempted to minimize bias, the possibility of subjective influence cannot be entirely eliminated. In addition, it will be necessary to investigate the psychological acceptability of participants allowing their skills to be evaluated by AI rather than surgeons. Although this study collected participants' impressions of feedback comments, it did not verify the educational effects of real-time feedback or its transfer effects to the operating room. Further investigation is necessary to clarify the educational benefit of our skill assessment system. Nevertheless, we consider that our training system represents a promising pedagogical training tool, offering real-time quantitative and objective feedback to trainees, and could enhance trainees' motivation, and thus shorten their learning curve. This will facilitate effective training, even in Japan's current surgical environment of surgeon shortages and restricted working hours. These improvements are expected to enhance medical safety, increase efficiency, and reduce surgeons' workload. Using the participants' feedback from this investigation, we intend to revise the feedback contents more user-friendly. As a preliminary step to validation in actual surgery, we plan to first establish benchmarks in wet-lab training using the cadaveric pigs described in this paper and then examine the correlation through cadaveric training. We have already integrated the present Mocap system into cadaveric laparoscopic training, and plan to examine the correlation between the two training programs using various parameters. In addition, our future work will include verifying the learning curve through continuous measurements of the same participants, as well as verifying the performance of trainees who have utilized this system in cadaver training and actual surgery by analyzing surgical instrument movements. Ultimately, we aim to expand our surgical competency assessment framework [16].

Conclusion

Our real-time feedback system was viewed positively by participants, with the ML algorithm demonstrating particular reliability in GOALS score prediction. This system can provide objective feedback to trainees, enhancing simulation-based training efficacy and potentially accelerating surgical skill acquisition.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00464-025-12018-3.

Funding This work was supported by JSPS (Japan Society for the Promotion of Science) Grants-in-Aid for Scientific Research (C) (JP17K08897), (A) (JP18H04102), (B) (JP21H00893), (A) (JP23H00480), and (B) (JP24K00443), a Grant-in-Aid for Challenging Research (Exploratory) (JP23K18486), Grant-in-Aid for JSPS Fellows (JP22KJ0118), and AMED (Japan Agency for Medical Research and Development) under Grant Number JP22vk0124006.

Declarations

Conflict of interest Drs. Koki Ebina, Takashige Abe, Kiyohiko Hotta, Yan Lingbo, Chihiro Kamijo, Madoka Higuchi, Masafumi Kon, Hiroshi Kikuchi, Haruka Miyata, Ryuji Matsumoto, Takahiro Osawa, Sachiyo Murai, Taku Senoo, Shunsuke Komizunai, Teppei Tsujita, Kazuya Sase, Xiaoshuai Chen, Atsushi Konno, and Nobuo Shinohara have no conflicts of interest or financial ties to disclose.

Ethical approval This study was approved by the Ethical Review Board for Life Science and Medical Research of Hokkaido University Hospital (No. 018–0257). All procedures performed in studies involving human participants were in accordance with the ethical standards of the Institutional Research Committee and the Declaration of Helsinki of 1964 and its subsequent amendments or comparable ethical standards.

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